

## **Project Name**

### **Global Logistics Intelligence Platform (GLIP)**

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#### **1. Project Background**

Global Logistics Company (GLC) is an international logistics provider that transports goods across the GCC, Middle East, Europe, and Asia using road, air, sea, and rail transportation.

As the business expanded, management noticed increasing transportation costs, more delivery delays, and inconsistent carrier performance. Operational data is currently spread across multiple systems, making it difficult to monitor performance and identify the root causes of inefficiencies.

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#### **2. Business Problem**

The company is experiencing:

- Increasing transportation costs.
  - Higher delivery delays.
  - Inconsistent carrier performance.
  - Limited visibility into logistics operations.
  - No centralized dashboard for decision-making.
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#### **3. Project Goal**

Develop a centralized Business Intelligence dashboard that enables management to monitor logistics operations, identify performance issues, analyze operational trends, and support data-driven decision-making.

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#### **4. Business Objectives**

The dashboard should help the company:

- Improve the On-Time Delivery Rate.
- Reduce transportation and freight costs.

- Identify the main causes of shipment delays.
  - Evaluate carrier performance.
  - Improve route efficiency.
  - Monitor logistics KPIs in real time.
  - Support strategic and operational decision-making.
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**5. Stakeholders**

| Stakeholder        | Responsibility       | Main Interest                |
|--------------------|----------------------|------------------------------|
| CEO                | Executive Management | Overall business performance |
| COO                | Operations           | Logistics efficiency         |
| Logistics Manager  | Logistics Operations | Delivery performance         |
| Finance Manager    | Financial Control    | Cost and profitability       |
| Operations Manager | Daily Operations     | Shipment monitoring          |
| Sales Manager      | Customer Service     | Customer performance         |

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**6. Scope**

**Included**

- Shipment performance
- Carrier performance
- Customer analysis
- Route analysis
- Cost analysis
- Vehicle utilization
- Warehouse performance
- Weather impact

- Driver performance
- Executive KPIs

### **Excluded**

- HR management
  - Payroll
  - Inventory management
  - Procurement
  - Marketing analytics
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## **7. Key Business Questions**

Management wants answers to questions such as:

### **Executive**

- How many shipments have been completed?
- What is the On-Time Delivery Rate?
- What is the total revenue?
- What is the total freight cost?
- What is the profit margin?

### **Operations**

- Which carriers have the highest delay rates?
- Which routes are the most expensive?
- Which transport mode performs best?
- Which warehouses have the highest delays?
- Which drivers have the best performance?

### **Customer**

- Who are the top customers?
- Which industries generate the highest revenue?

- Which regions have the most customers?

#### **Finance**

- What is the cost per shipment?
- What is the average freight cost?
- Which shipments are least profitable?

#### **Sustainability**

- Which transport mode produces the highest CO<sub>2</sub> emissions?
  - How does fuel consumption vary by transport mode?
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### **8. Success Criteria**

The project will be considered successful if it enables management to:

- Monitor all logistics KPIs from one dashboard.
  - Identify delay patterns quickly.
  - Compare carrier performance.
  - Analyze transportation costs.
  - Improve operational efficiency.
  - Support faster and better business decisions.
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### **9. High-Level KPIs**

#### **Delivery KPIs**

- Total Shipments
- Delivered Shipments
- Delayed Shipments
- On-Time Delivery Rate
- Average Delay Hours

#### **Financial KPIs**

- Total Revenue
- Total Freight Cost
- Profit
- Profit Margin
- Cost per Shipment

#### **Operational KPIs**

- Average Transit Time
- Average Distance
- Vehicle Utilization
- Warehouse Utilization
- Route Efficiency

#### **Sustainability KPIs**

- Total Fuel Consumption
- CO<sub>2</sub> Emissions
- Green Transport Rate

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### **10. Expected Deliverables**

- Business Requirements Document (BRD)
- Data Dictionary
- Entity Relationship Diagram (ERD)
- Star Schema Data Model
- Python-generated synthetic dataset
- SQL analysis scripts
- Tableau dashboards
- Executive presentation with insights and recommendations